

System Architecture: Autonomous Astronomical Anomaly Discovery Platform

Data Ingestion Layer

CSV Upload

NASA JPL API

ZTF/LSST Stream

WebSocket Ingest

Processing Layer

Preprocessing & Normalization

Sliding Window (W=32)

Deep Learning Layer

Autoencoder
(64→16→64)

VAE
(64→32→ μ, σ →32→64)

Transformer
(d=32, h=4, L=2)

Agentic AI Layer

Ensemble
Fusion

SLM
Reasoning

RL Policy
Optimizer

Vector
Search

Alert
Dispatch

Persistence & UI Layer

SQLite
Database

Provenance
Ledger

JWT Auth &
Rate Limiting

React Frontend
& Citizen Portal